

Conditionals: first steps

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5 September 2017

1. Background

- **Sentences** versus **propositions**

- Propositions: things that can be true or false; things we mean, assert, deny, believe, are more or less confident of...

- For the proposition that snow is white to be true is for snow to be white

- **Context sensitivity.** A context-sensitive sentence is one that can be used, literally, to express multiple propositions.

- **Connectives.** In formal languages, we find expressions called *connectives*, each of which combines with a certain specified number of sentences to form a new sentence.

- Familiar unary connectives: $\neg$, $\diamond$, $\Box$

- Familiar binary connectives: $\wedge$, $\vee$, $\supset$

The meaning of an n -ary connective can be specified as a *function* from n -tuples of propositions to propositions. A context-insensitive connective is associated with a single such function. A context-sensitive one can be associated with different functions on different occasions.

- A **truth-functional** connective is one that expresses a function f from n -tuples of propositions to propositions with the following property: whenever P_1 and Q_1 are either both true or both false, and ... and P_n and Q_n are either both true or both false, then $f(P_1, \dots, P_n)$ and $f(Q_1, \dots, Q_n)$ are either both true or both false. Not all connectives are truth functional: for example, even when ϕ and ψ are both true, $\Box\phi$ may be true and $\Box\psi$ false.

2. Conditionals

Our central focus: declarative English sentences of the following forms, and their meanings:

If ϕ , ψ If ϕ , then ψ ψ if ϕ

In each case we have an **if-clause**, 'if ϕ ', and a **main clause**, ' ψ ' (or 'then ψ ')

(Some authors use 'conditional' for the sentences; others, for the propositions they express.)

However it is worth paying some attention to the various ways in which similar meanings can be expressed without ‘if’:

- (1) a. Unless there has been an accident she is nearly home
- b. Had we not won, we would have been in trouble
- c. Whatever Bill says, Mary will quit her job
- d. One more such victory and we are lost
- e. My kids would love a pet dog

We should also not forget about conditional questions and imperatives:

- (2) a. Where would you live if you were a millionaire?
- b. If the butler did it, what weapon did he use?
- c. If you’re so smart, why aren’t you rich?
- d. Attack immediately if you spot the enemy.
- e. If you’re so smart, prove it yourself.

3. ‘If’ as a binary connective

There is a long tradition of treating ‘if’ as an expression of the same kind as ‘and’ or the ‘ $\wedge$ ’ of formal logic. On the obvious way of implementing this idea, we would expect the proposition expressed by ‘If ϕ , ψ ’ in a given context c to be the result of applying the function expressed by ‘If’ in c to the ordered pair of the proposition expressed by ϕ in c and the proposition expressed by ψ in c : $\llbracket \text{If } \phi, \psi \rrbracket^c = \llbracket \text{If} \rrbracket^c(\llbracket \phi \rrbracket^c, \llbracket \psi \rrbracket^c)$.

This idea faces several problems, some deeper than others.

Problem 0: Sometimes, including a context-sensitive sentence in a conditional makes it natural to interpret it differently from how it would have been interpreted if it had been uttered all by itself:

- (3) If the White House basketball game took place yesterday, no tall people were at the meeting.

Response: remember that “contexts” here are theoretical entities. Distinguish *what sentence ϕ expresses in the context of utterance u* from *what would have been expressed by an utterance of ϕ if one had taken place instead of u* .

Problem 1: ‘If ϕ , ψ ’, the ‘if’ clause ‘if ϕ ’ is itself a meaningful constituent—an adverbial clause. (Compare: in ‘I left because it was late’, ‘because it was late’ is a meaningful constituent, whereas ‘I left because’ is just a string of words.)

We can get around this problem by taking ‘if ϕ ’ to have the meaning of a unary operator—a function from propositions to propositions—and ‘if’ to have the meaning of a function from propositions to such functions.

Problem 2: in

- (4) If I resigned tomorrow, I would be hired by Google next week

it’s quite unclear how to answer the question what propositions are expressed in the context by the constituent sentences ‘I resigned tomorrow’ and ‘I would be hired by Google next week’. The former seems like nonsense; the latter is not nonsense, but it’s got a kind of context-sensitivity that doesn’t show up in the original sentence. What we’d like to say is that the proposition expressed by (**), as uttered by me now, is determined by applying a certain function to the proposition that I *will* resign tomorrow and the proposition that I *will* be hired by Google next week.

Response: all right. What we are really interested in here is a certain way of building one sentence out of two others, which isn’t just a matter of combining them with the word ‘if’, but also sometimes involves making changes of tense and inserting ‘would’. Our task is to explain how the meaning of the output of this operation depends on the meanings of its inputs.

Problem 3: What are the two propositions that would be inputs to the relevant function in a sentence like

- (5) If a programmer lives with non-programmers, the programmer is frustrated

Certainly the main clause does not contribute the proposition that the one and only programmer (in the world, or in some limited domain) is frustrated, or that some particular person is frustrated, or anything like that! The question seems misconceived!

This issue is actually quite pervasive, especially with ‘does-will’ conditionals.

- (5a) If Jack sees Jill next week, he will wave

What proposition would play the role of the consequent here? Not just the proposition that Jack will wave sometime or other, or the proposition that Jack will wave next week, or even the proposition that Jack will wave at Jill next week...

Let’s come back to cases like these many weeks from now (when we look at Kratzer’s theory). For now we will try to focus on cases where it’s clear that we are dealing with two specific propositions.